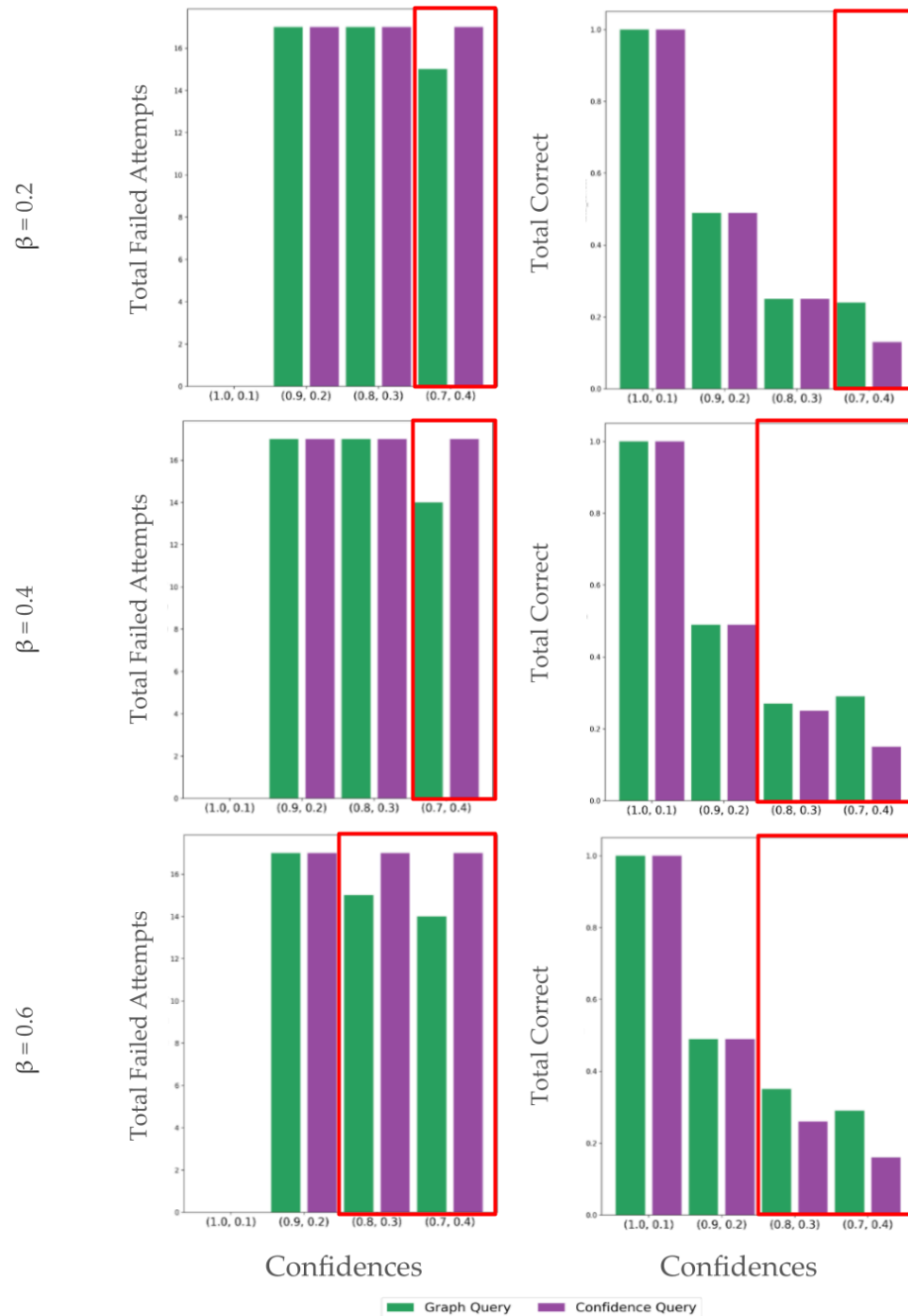


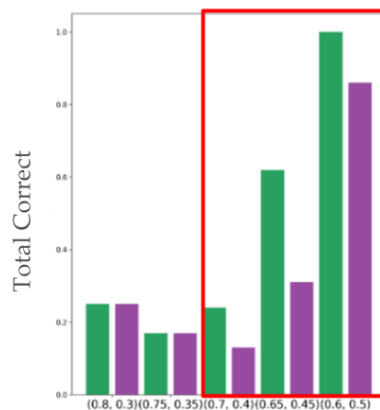
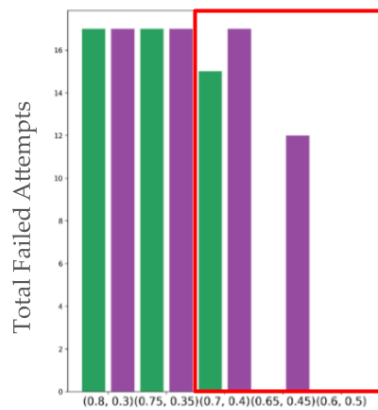
Rebuttal Supplementary Materials

GraphQuery v. ConfidenceQuery: Simulation Results

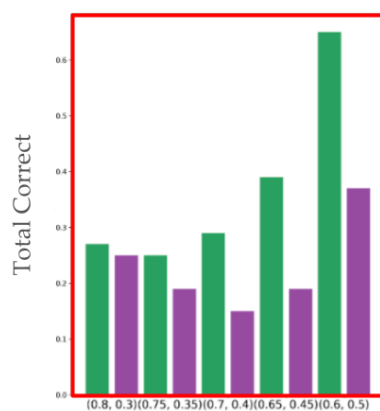
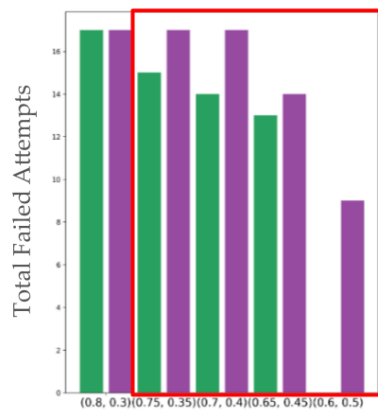
Uniform random noise in each module's query cost q_i
 $q_i \sim \text{Uniform}[(1-\beta)Q, (1+\beta)Q]$, $Q = 0.32$



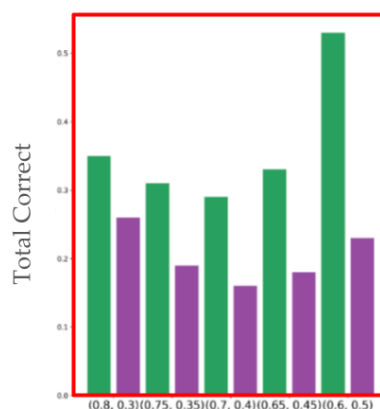
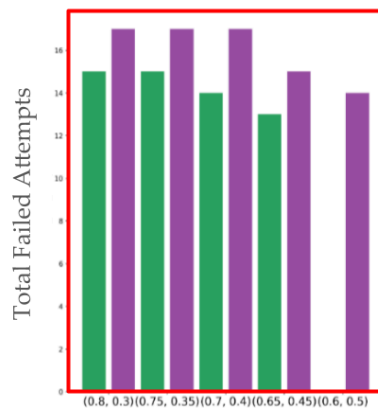
$\beta = 0.2$



$\beta = 0.4$



$\beta = 0.6$



Confidences

Confidences



GraphQLy v. ConfidenceQuery: Feeding Study Results

Subjective scores (top) and objective scores (bottom) for N=5 users without mobility limitations.

Means	Mental/Physical Demand ↓	Effort ↓	Subjective Success ↑	Satisfaction ↑
GraphQLy	1.8	2	4.2	3.8
ConfidentQuery	3.4	3.6	2.2	1.6
Means	Mean Queries per Bite ↓	Mean Executions per Bite ↓	Mean Successful Bites per Trial ↑	
GraphQLy	2.32	1.68	0.92	
ConfidentQuery	6.28	1.12	0.44	

References

- [1] Jenamani, R.K., Silver, T., Dodson, B., Tong, S., Song, A., Yang, Y., Liu, Z., Howe, B., Whitneck, A. and Bhattacharjee, T., 2025. FEAST: A Flexible Mealtime-Assistance System Towards In-the-Wild Personalization. Robotics: Science and Systems (RSS), 2025.
- [2] A. Padmanabha, J. Gupta, C. Chen, J. Yang, V. Nguyen, D. J. Weber, C. Majidi, and Z. Erickson, “Independence in the home: A wearable interface for a person with quadriplegia to teleoperate a mobile manipulator,” in Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction, 2024, pp. 542–551.
- [3] Z. Wu, B. Ai, T. Silver, and T. Bhattacharjee. 2025. SAVOR: Skill Affordance Learning from Visuo-Haptic Perception for Robot-Assisted Bite Acquisition, in Conference on Robot Learning (CoRL), 2025.
- [4] Xia, K., Jiang, H., Man, Z., Jiang, Y. and Yao, Z., 2024. LC-Dual: Coupling predictive information and redundant strategies for autonomous vehicle lane change trajectory planning. IEEE Transactions on Intelligent Vehicles.
- [5] Gosala, N., Bühler, A., Prajapat, M., Ehmke, C., Gupta, M., Sivanesan, R., Gawel, A., Pfeiffer, M., Bürki, M., Sa, I. and Dubé, R., 2019, May. Redundant perception and state estimation for reliable autonomous racing. In 2019 International Conference on Robotics and Automation (ICRA) (pp. 6561-6567). IEEE.
- [6] Huang, B., Wang, Y., Yang, X., Luo, Y. and Li, Y., 2024. 3d-vitac: Learning fine-grained manipulation with visuo-tactile sensing, in Conference on Robot Learning (CoRL), 2024.
- [7] X. Ma, S. Patidar, I. Haughton, and S. James. 2024. Hierarchical Diffusion Policy for Kinematics-Aware Multi-Task Robotic Manipulation. CVPR (2024).
- [8] Rubenstein, M., Cornejo, A. and Nagpal, R., 2014. Programmable self-assembly in a thousand-robot swarm. Science, 345(6198), pp.795-799.